

NISHAN MAHARJAN

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PROFILE

Final-year Computer Science student specializing in AI, Data Science, and Computer Vision, with hands-on experience training and deploying models: a custom-trained YOLOv8 detection pipeline for real-time sports analytics, and classification/clustering models for customer churn prediction. Comfortable owning a project end to end, from data and model through to a working, deployed application.

SKILLS

- **AI / Machine Learning:** PyTorch, Scikit-learn, YOLOv8 (custom object detection), ByteTrack, OpenCV, EasyOCR, Computer Vision, Logistic Regression, K-Means Clustering, model evaluation (mAP, precision/recall/F1)
- **Backend & APIs:** FastAPI, MongoDB (Motor, Atlas), Pydantic, REST API design, JWT authentication, OAuth
- **Data & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Jupyter Notebook, Google Colab
- **Frontend & Tools:** React (Vite), JavaScript, HTML, CSS, Figma, Git/GitHub, Docker, ffmpeg
- **Soft Skills:** Communication, team collaboration, problem-solving, adaptability

TRAINING & CERTIFICATIONS

- AI and Data Science Training, Techaxis (3 months, 2026)
- Data Analysis Workshop (Python & Power BI), Vrit Jobs
- ML Model / AI Automation Workshop (2026)

KEY TECHNICAL PROJECTS

FutStats: Computer-Vision Futsal Analytics Platform 2026, 2nd Runner-Up at FutStats Hackathon

- Trained a custom YOLOv8 model (mAP@50: 92%, player + ball classes) on a self-built dataset of 498 hand-labeled frames (3,738 annotations) from own match footage.
- Tracked players with ByteTrack; identified and quantified ID fragmentation under camera motion (~9x ID-inflation on panning clips) and kept robust team-level stats (possession, heatmaps) as the headline numbers rather than hiding the limitation.
- Used K-Means clustering for automatic team assignment from kit colour; built as part of a team within a 48-hour hackathon window.

Stack: Python, YOLOv8, ByteTrack, OpenCV, scikit-learn, FastAPI, ffmpeg

Web Based Expense Tracker and Decision Support System 2026

- Full-stack app with a two-stage OCR pipeline (OpenCV preprocessing, then EasyOCR/PyTorch extraction, then rule-based parsing) and confidence scoring to flag unreliable extractions.
- Built budget-vs-actual tracking, a trip-budget tracker, and a Plotly/pandas analytics dashboard (category trends, spending-pace projection).
- Owned the FastAPI backend (4 routers), MongoDB schema, JWT/Google OAuth authentication, and Docker development setup end to end.

Stack: FastAPI, MongoDB (Motor), React 19, EasyOCR, OpenCV, PyTorch, Docker

Telco Customer Churn Analysis 2026

- Built a Logistic Regression churn classifier (79.3% accuracy, 57.3% F1) and a K-Means segmentation model identifying 4 customer segments, including an "At-Risk New High-Spenders" group with 48.3% churn.
- Structured the pipeline with OOP (dedicated classes for preprocessing, feature engineering, modeling, clustering) and an interactive ipywidgets dashboard.

Stack: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

OTHER PROJECTS

- **Quote Generator App (2024):** Responsive HTML/CSS/JS app with category selection and dark/light mode, hosted on Netlify.
- **UI/UX Prototyping (College):** Multiple Figma prototypes translated into working front-end interfaces.
- **Exploratory Data Analysis (College):** Pandas/NumPy-based EDA on real-world datasets.
- **Panel Discussion (2024):** Organized a panel on the role of IT students in energy-management innovation.

EDUCATION

Bachelor in Computer Science 2023 – Present

IIMS College, Kathmandu (affiliated to Taylor's University, Malaysia)

Plus 2 in Science 2021 – 2023

Trinity International College, Kathmandu, GPA: 3.49 (A)

SEE 2014 – 2021

Nepal APF School, Kathmandu, GPA: 3.85 (A+)